enable code to run on computing targets such as GPUs. Nx is a low-level library that's leveraged by higher-level libraries to implement common machine learning patterns.

Bumblebee[9] is a library that leverages Nx to execute pretrained neural network models. It works out of the box with HuggingFace models, which is the most popular host for machine learning models. Bumblebee has been shown to work with popular models such as "Whisper" and "Stable Diffusion."

Livebook[10] isn't strictly machine learning related, but it's commonly used in a similar way to Jupyter notebooks, which are used in the Python community to share executable code alongside the results of the code. Livebook executes Elixir code located in cells and creates a formatted, documented notebook of code that can be shared between team members.

The development of machine learning capabilities in Elixir is incredibly exciting. Elixir's capabilities with robust data pipelines and distributed code execution make it a perfect contender in the machine learning space because of the amount of work that goes into bringing machine learning models to production. However, there's still a lot of work to do in this space. Keep an eye on it because it has shown so much promise so far.

For our final topic, let's look at potential type system support in Elixir.

Type Systems

For better or worse, one of the most commonly asked questions by developers coming from another language is whether Elixir provides types support. The common argument is that static types defined in an application allow for more predictability and guarantees about how an application will perform. (Whether those claims are true or not is a matter of opinion.)

Many Elixirists take advantage of pattern matching and structs to achieve a loose version of typing. If you use a struct match like `message = %SmsMessage{}` in your function head, then you know that `message` will be an `SmsMessage`. This actually goes a long way, but it's not a true types support.

Dialyzer is an Erlang package that provides a "success typing" implementation for BEAM languages. There is an Elixir library that wraps it called Dialyxir.[11] This package can catch a lot of logic mistakes in your code, but it's not native

9. https://github.com/elixir-nx/bumblebee
10. https://livebook.dev/
11. https://github.com/jeremyjh/dialyxir